

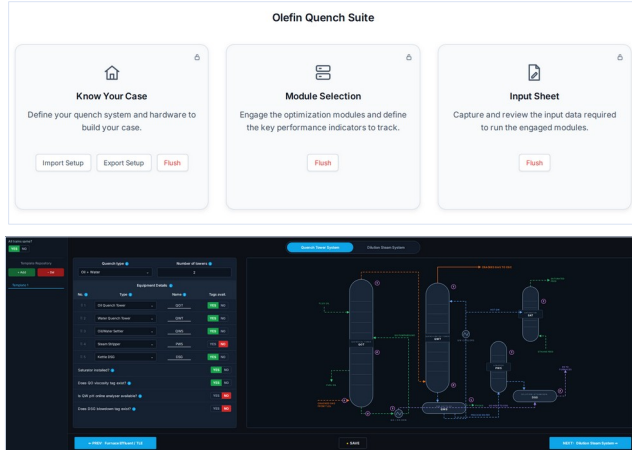
PRODUCT OVERVIEW

QuenchX360 is a decision-support application for the olefin quench system. Domain and ML agents are configured on the unit's own historian and sensor tag data, covering the quench oil tower, quench water tower, settler, process water stripper, dilution steam generator (DSG) and feed saturator. It tracks quench oil viscosity, flux oil demand, pH / emulsion health, tower fouling, heat recovery and DSG / saturator performance. Output reaches the process engineer as ranked, quantified actions, each reported against the parameter that caused it. Every model agent is configured, validated and commissioned per site by Ingenero process engineers, so the deployment reflects that unit's configuration and operating envelope.



01 Unit Configuration

Shown: Olefin quench suite – interactive configuration, modules and sensor entry

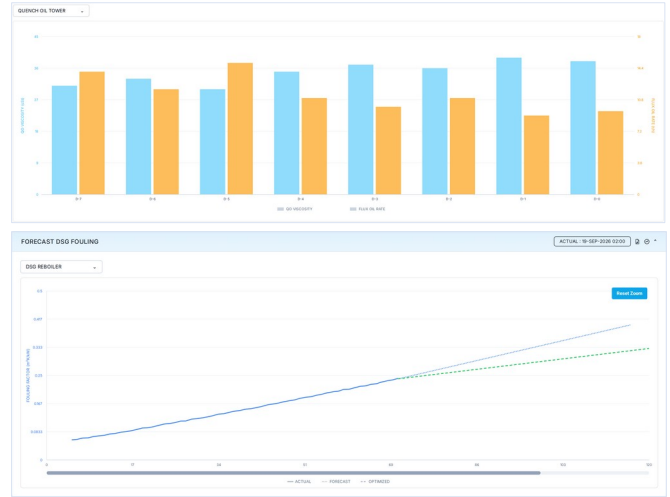


- Know Your Case** Configure quench oil & water towers, settler, stripper, DSG and saturator
- Module Selection** After configuring the quench system, select the modules – Domain & AI agents
- Input Sheet** Based on configuration & Module selection – relevant sensor info needs to be filled
- Interactive Config** Quench oil loop, quench water loop, DSG & saturator interactive PFD available



02 Plant Data and Model Execution

Shown: QO viscosity vs flux oil, DSG fouling forecast current vs optimum.



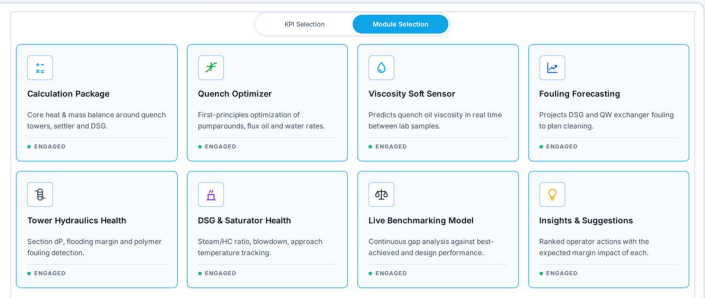
- Inputs** Sensor tag entry with design tag values and lab analysis (viscosity, pH)
- Processing** tag validation, steady state framing and feed / severity regime detection
- Outputs** predicted KPIs, ranked actionable with cause, trends, alerts



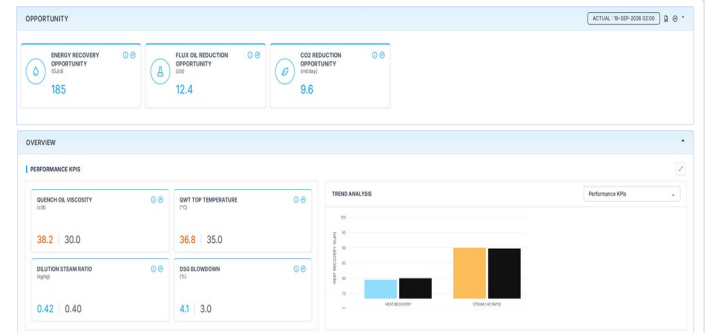
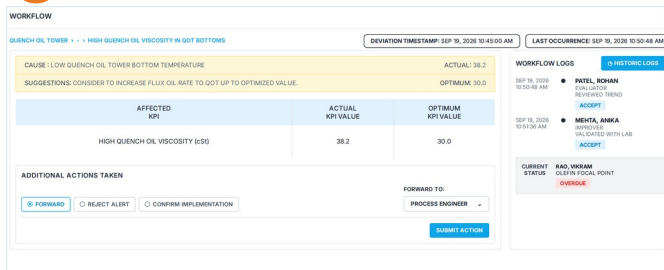
03 AI-Powered Model Agent Suite

Agent/Module selection shared along with detailing of the quench specific details

- Quench Oil Viscosity** Soft sensor predicts QO viscosity and optimizes flux / wash oil rate.
- Tower Hydraulics & Fouling** Tracks section-wise dP and flags polymer fouling early in QOT/QWT.
- Emulsion & pH Health** Monitors QW pH, settler interface and emulsion risk with warnings.
- Heat Recovery** Maximizes quench heat recovered to dilution steam and process users.
- DSG & Saturator Performance** Tracks steam/HC ratio, DSG fouling, blowdown and saturator approach.
- Process Water Stripper** Estimates stripping efficiency and dissolved HC carryover to DSG.
- Fouling Precursor** Soft sensor estimates styrene/indene precursors driving QW fouling



04 Opportunity and Operator Actions



+3-5%

Quench heat recovery
from QO / QW pumparound optimization

10-15%

Flux oil reduction
viscosity soft sensor based control

+20%

DSG exchanger run length
fouling forecast based cleaning

Basis: optimization respects quench, furnace dilution steam and CGC suction limits; suggestions take care of step change for smooth implementation.



05 Platform

Historian connectivity

plug and play with leading plant historians

Simulation integration

third party simulators integrate without custom work

Daily reporting

downloadable daily reports with interactive trends

Gen AI assistant

answers plant questions in the operating workflow

Build your own dashboard

engineers build the views their unit needs

DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

FLEXIBILITY

Oil or water quench; with/without saturator; selectable KPI set

SCALE

Repeat sites reuse the configuration, with no/minimal rework per plant.

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